

DGE Exploration: Technical Case Study

Scientist-Facing Exploration of Differential Gene Expression — Full Reasoning Chain

This is the condensed technical companion to the portfolio summary. It walks the complete reasoning chain — discovery, data, product, design, and build plan — in one document. For the individually numbered source documents (00 – 14), see the docs/ folder.

1. Approach and Mental Model

The starting point is an existing DGE (differential gene expression) result — a table of roughly 22,000 genes produced by a DESeq2-based RNA-seq analysis comparing dexamethasone-treated and untreated human airway smooth muscle cells. The exercise treats this as a real FDE (Forward Deployed Engineer) engagement: a pharmaceutical research team hands over an existing analysis output and needs a way for scientists to explore it.

The whole project follows one operating loop:

CLIENT → existing scientific workflow → existing data output → understand data and its context → ask what scientists need to do → identify friction → translate friction into product needs → build prototype → test with users → iterate

Every section below is one pass through this loop, applied progressively — from questions, to data, to product, to design, to a buildable implementation plan.

2. Discovery: What Should Get Asked Before Touching the Data

In a real engagement, this section would come first *and* would happen through conversation, not analysis. Since there’s no client to interview in this exercise, this section documents what a real discovery phase would have covered, organized around the same six-stage sequence the rest of the project follows: **Client Context** → **Data** → **User Workflow** → **Requirements** → **Prototype** → **Evaluation**.

Stage	Key questions	What goes unanswered if skipped
Client Context	What does success look like, concretely? What’s already been tried? Who is this actually for?	Builds the technically impressive thing instead of the used thing
Data	Where does authoritative experiment metadata live? Is this one comparison of several? Are there QC flags or batch effects?	The “experiment overview” screen gets built on unverified assumptions
User Workflow	What annotation sources do scientists already trust? What’s the next action after flagging a gene?	Risk of building a redundant tool, or one that dead-ends at the moment of insight
Requirements	Which job are scientists actually doing — target ID, compound comparison, mechanism exploration? How much trust does a number need to earn?	The wrong exploration lens gets prioritized; traceability gets under- or over-built
Prototype	Are there compliance constraints on where data can live? Who needs to approve the interface?	Screens get designed around infrastructure that doesn’t exist
Evaluation	How would you know this actually changed how someone works?	“Success” stays untestable after launch

The practical argument: a five-minute client conversation at the Client Context stage would likely have answered, immediately, several things that took multiple passes of data analysis to approximate in Section 3 below. Data analysis here is a substitute for a conversation that doesn’t exist, not a replacement for having it.

3. Data Analysis and Reasoning

What the dataset actually contains

The DGE table has five measurements per gene: `baseMean` (average expression signal), `log2FoldChange` (direction and magnitude of change), `lfcSE` (uncertainty on that estimate), `pvalue` (statistical evidence), and `padj` (evidence after multiple-testing correction across ~22,000 simultaneous tests). None of these measurements is interchangeable, and none alone defines whether a gene is “important”:

- **Response magnitude is heavily concentrated near zero** — about 91.5% of genes fall between -0.5 and +0.5 log2FC, with only 74 genes above +2 and 109 below -2. A large background of small changes surrounds a much smaller population of larger responses.
- **Effect size and statistical evidence are genuinely independent dimensions.** Cross- tabulating absolute log2FC against `padj` (excluding missing values) shows 13,535 genes with small effects and weak evidence, but also 503 genes with only moderate effect (0.5-1 log2FC) and very strong evidence (`padj` < 0.001), and 138 genes with large effect (>2) and equally strong evidence. Ranking by fold change alone would bury the second group entirely.
- **`baseMean` is heavily right-skewed** — mean (≈972) roughly 9× the median (≈109) — so a linear-scale visualization would be dominated by a handful of highly expressed genes.
- **`padj` is missing for 17.7% of genes** (3,891 of 22,008). Missing must never silently become “not significant” — that’s a different claim than “no measurement available.”

The reframing this produces

Initial framing: *scientists have 22,000 genes and need to find the interesting ones.* After inspecting the data, the more accurate framing is: *scientists are navigating multiple independent dimensions at once (expression, effect, uncertainty, evidence), and the challenge isn’t finding a row — it’s knowing which region of a multidimensional result deserves attention.* That shift is what turns “build a searchable table” into “build an exploration environment,” and it recurs as the FDE reasoning pattern throughout this project: *question → why does this matter? → inspect the data → what did we learn? → what product question does that create?*

4. Scientist Workflow

Modeling what a researcher actually does with a DGE result, stage by stage, surfaces where today’s typical tooling (spreadsheets, static plots, ad hoc scripts) creates friction:

Stage	Scientist’s question	Where today’s tools create friction
Orientation	What am I looking at?	Raw tables don’t communicate experimental context
Understand response	What happened overall?	22,000 rows don’t summarize themselves
Notice patterns	What stands out?	Requires manually adjusting thresholds and comparing columns
Explore a pattern	Why is this interesting?	Comparing effect + evidence + expression means switching between tools
Narrow to a subset	Which genes are here?	Filter criteria get lost once a subset is exported
Investigate a gene	What’s happening with this gene?	Statistical and biological context live in separate places
Add biological context	What might this mean?	Fragmented across annotation databases, pathway tools, literature
Decide on follow-up	What’s worth investigating next?	Evidence has to be manually assembled before a decision can be made

The central design principle that follows: **the scientist should never be assumed to already know which gene they’re looking for.** The product needs to support movement from overview → pattern → subset → gene → context, not just search.

5. Product Requirements and Feasibility

What the product should do — and explicitly not do

The core requirements, translated from the workflow above: provide an immediate dataset overview; surface the overall

response landscape; support multiple simultaneous exploration lenses (magnitude, direction, evidence, expression, uncertainty) rather than one composite score; preserve the underlying measurements rather than replacing them with opaque categories; support progressive narrowing from pattern → subset → gene; avoid presenting any “most important genes” ranking without a stated, visible basis; keep missing values explicitly distinguishable from weak evidence; and treat gene search as a secondary interaction, not the primary way in.

Checking requirements against what the data can actually support

For every requirement, the honest question is: can this be built from the five columns already available (**directly supported**), does it need to be derived through calculation (**derived**), or does it require data the client hasn’t provided (**requires additional data**)?

Fully supported by the current dataset: experiment/response overview, pattern discovery, multidimensional exploration, dynamic gene subsets with preserved filter criteria, and gene-level statistical detail.

Requires additional data the client would need to supply: gene symbols and descriptions, pathway membership, Gene Ontology terms, literature connections, drug-target status, disease associations, multi-treatment or dose-response comparison, and sample-level expression data (needed for anything like a heatmap, since the DGE table itself has no sample × gene matrix).

This gap list is exactly what turns into real discovery questions in a live engagement (Section 2) — “what annotation source do you already trust,” “is this one comparison of several” — rather than being treated as a fixed technical limitation.

6. Application Design: Flow and Screens

The interaction flow

```
Receive DGE Results → Orient to Experiment → Understand Overall Response →  
Notice Potentially Informative Patterns → Explore a Pattern →  
Narrow to a Subset → Investigate Individual Genes → Add Biological Context →  
Decide What Deserves Follow-Up
```

Each stage needs to preserve context from the one before it — if a gene was reached via “large negative responses, $\text{padj} < 0.05$,” that provenance should stay visible at the gene level, not get lost the moment the researcher clicks through.

The five core screens

1. **Experiment Overview** — orientation only, not analysis: treatment, comparison, system, gene count, and what each measurement means.
2. **Response Landscape** — the first analytical screen: effect and evidence distributions, plus a set of neutrally-worded “potentially informative region” cards (e.g. “Large negative responses — 109 genes”) that the researcher chooses to explore, rather than a single ranked “top genes” list.
3. **Pattern Exploration** — an interactive workspace where filter criteria (e.g. $\log_2\text{FC} < -1.5$, $\text{padj} < 0.05$) update the gene count live, with the current criteria always visible so filtering never becomes opaque.
4. **Gene Set** — a sortable table of the resulting subset, with the pattern and criteria that produced it still attached.
5. **Gene Detail** — full statistical evidence for one gene, the selection path that led there, and a reserved (initially empty) section for biological context, clearly separated from the experimental evidence above it.

The design principle running through all five: **organize around what the scientist wants to know next, not around what can be charted from a given column.** The homepage is deliberately not a volcano plot with a search bar — it’s an orientation screen, because that’s the question a scientist actually asks first.

7. Technical Specification

Stack and architecture

The prototype is intentionally **frontend-first**: React, TypeScript, Vite, Plotly or D3 for visualization, React state/Context for application state, and a local JSON dataset — no backend, database, or authentication required for V1. The reasoning: the product question being tested is about scientific interaction, not infrastructure, so the architecture should prioritize iteration speed over completeness. A backend and an AI service are both explicitly deferred, but the architecture is intentionally left with room to add them later (`Frontend → API → {Dataset, Analysis, Annotation} services`) without redesigning the interaction model.

The application is organized into five layers — UI, Interaction, State, Data, and the raw Dataset — with one governing rule: **state lives at the lowest level that needs to share it, but no lower.** Global state covers experiment, comparison, filters, and

selected genes; everything else (tooltip visibility, panel expansion) stays local.

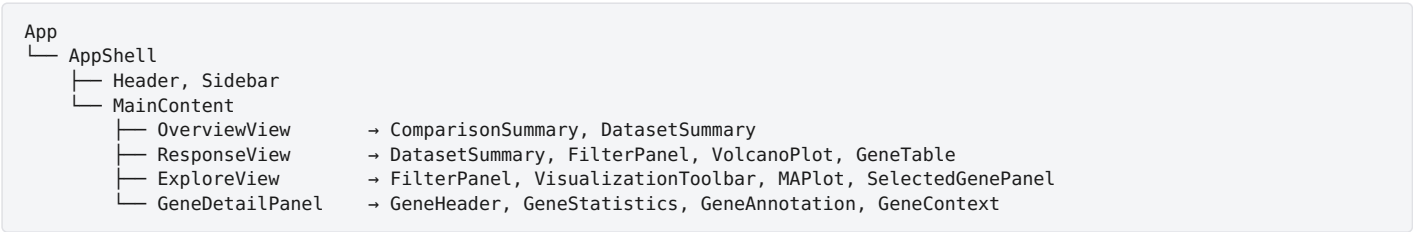
Data contract

Every gene record follows a minimum schema:

Field	Type	Required	Supports
gene_id	string	Yes	Unambiguous identification
gene_symbol	string	Yes	Human-readable display
log2FC	number	Yes	Direction and magnitude
pvalue	number	Yes	Unadjusted evidence
padj	number	Yes	Adjusted evidence used for filtering
baseMean	number	No	Expression context
annotation	string	No	Biological context, if available

Experiment and comparison metadata are modeled as separate objects from the gene-level results, and comparison direction must always be stated explicitly — “Treatment vs Control” and “Control vs Treatment” invert the meaning of the same log2FC value. Filter thresholds live in application configuration, not hardcoded in components, so a researcher can tighten or loosen criteria without ever modifying the underlying dataset. Five data principles govern all of this: preserve source evidence untouched; keep data separate from interaction state; make thresholds and classifications explicit rather than hidden; never fabricate missing biology (e.g. don’t build a heatmap without real sample-level expression data); and keep the frontend independent of whichever pipeline (DESeq2, edgeR, limma) produced the original analysis.

Component structure



Visualizations are reusable components that receive filtered data as props (`<VolcanoPlot data={filteredGenes} />`) rather than independently fetching or transforming data — and every visualization is also an interaction surface: clicking a point updates shared selection state, which simultaneously updates the gene table and detail panel. This is the single most important pattern in the whole build — a chart here is not a static image.

8. Build Plan

The implementation proceeds in eight milestones, each producing something runnable:

#	Milestone	Done when...
1	Project Setup	Dev environment runs locally
2	Data Layer	Real DGE dataset loads through a clean access layer
3	Application Shell	Navigation exists between the main views
4	Scientific Overview	Experiment context and summary statistics display correctly
5	Global DGE Exploration	Filtering, volcano plot, and gene table all read from one shared filtered dataset
6	Gene Investigation	Selecting a gene (from either plot or table) opens consistent, source-accurate detail
7	Cross-View Interaction	Selections and subsets survive switching between views
8	Refinement	Empty/error/loading states exist; values spot-checked against the source file

Milestone 5 carries a specific architectural test: if the volcano plot and gene table ever disagree on gene count because they’re running separate filtering logic, that’s an architecture defect, not a display bug — both must read from the same filtered dataset.

Definition of done for the MVP: open experiment → understand comparison → view overall response → filter results → inspect volcano plot → select a gene → inspect gene evidence → return to visualization — working reliably, end to end, with application values verified against the source DGE file. Everything beyond this (pathway analysis, expression heatmaps, external annotation, AI-assisted interpretation) is a deliberate, documented extension, not part of V1.

9. Closing Principle

One question governs every decision in this project, from the first data exploration pass to the last component spec: **does this make the scientist’s next analytical step easier?** The point was never to demonstrate how many technologies could be used — it was to take a real scientific workflow, understand the reasoning process behind it, and turn that understanding into software someone would actually want to open.